

Fuchs Philippe

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I am currently a PhD student in Biophysics and Quantitative biology at the Centre de Biologie Structurale (CBS), Montpellier, France. This project is under the joint supervision of [Luca Ciandrini](#) and [Marco Cosentino-Lagomarsino](#), respectively in the University of Montpellier and Milan. I will defend in october 2026 and plan to continue my research in academia.

Education

Current: Theoretical and experimental quantitative approaches to unveil the relation between resource allocation and cellular physiology

Centre de Biologie Structurale, Montpellier, France ([CBS](#))

- Cotutelle between the University of Montpellier ([CBS](#)) and the University of Milan, Italy ([IFOM](#))
- Supervisors : [Luca Ciandrini](#) and [Marco Cosentino-Lagomarsino](#)

A cell growth physics PhD centered around models and wet-lab work

Master Degree (2022): Master in Cell Physics ([Physique Cellulaire](#))

University of Strasbourg, France.

Master Thesis: Interdependence between mRNA Translation and Growth: the cost of unnecessary protein synthesis

Master diploma (2022): Magistère de Physique Fondamental ([MdPF](#))

University of Strasbourg, France.

Parallel master level degree followed from 2019 to 2022

Bachelor (2020): General Physics

University of Strasbourg, France.

Specialisation in mathematics ([MPA](#)).

Internships

2022 - Interdependence between mRNA Translation and growth : the cost of unnecessary protein synthesis

Centre de Biologie Structurale ([CBS](#)), Montpellier, France.

M2 internship under the supervision of [Luca Ciandrini](#)

Path integrals in quantum mechanics

Institut Pluridisciplinaire Hubert Curien ([IPHC](#)), Strasbourg, France

Bachelor internship under the supervision of [Janos Polonyi](#)

Data Analysis of a Cosmic Ray Detector [KASCADE](#)

Institut Pluridisciplinaire Hubert Curien ([IPHC](#)), Strasbourg, France

Bachelor internship under the supervision of [Eric Baussan](#)

Skills

Programming Python (Pandas, PyTorch, NumPy, etc.), ROOT, C++, Maple.

Mathematics Additional math course followed along my main education in general and cell physics (([ITI IRMIA++](#)), [MPA](#))

Wet-Lab Cell cultures in micro-reactors and plate readers (*E. coli*, RNA, Fluorescence and protein measurements, Growth rates measurements)

French Native proficiency

Dutch Native proficiency

English C1 proficiency - Cambridge English Language Assessment

German B1 proficiency - High School Diploma

Spanish A2 proficiency - High School Diploma

Miscellaneous ImageJ, SurfaceEvolver, Linux, $\text{\LaTeX}$ (Overleaf), Microsoft Office, Git.

Events

CONFERENCE CONTRIBUTIONS (TALK OR POSTER)

Sept. 2024	Conference , Physics of Living Matter (Centuri) - Poster	Marseille, France
June. 2024	Conference , International Physics of Living Systems (IPoLS) - Talk	Trieste, Italy
Oct. 2023	Conference/School , Advanced School on Quantitative Principles in Microbial Physiology (ICTP) - Talk	Trieste, Italy
June 2022	Conference , International Physics of the Living System (iPoLS) - Poster	Montpellier, France

SCHOOLS AND OTHER EVENTS

July 2023	Summer School , Economic Principles in Cell Physiology (EPCP)	Paris, France
May 2023	Conference , Modeling in Ecology an Evolution (MEE)	Montpellier, France
July 2022	Summer School , Economic Principles in Cell Physiology (EPCP)	Paris, France
2021	Competition , French Physicist Tournament (FPT)	Strasbourg, France

Publications

Incoherent feedback from coupled amino acids and ribosome pools generates damped oscillations in growing bacteria
 Droghetti, R., **Fuchs, P.**, Iuliani, I., Firmano, V., Tallarico, G., Calabrese, L., Grilli, J., Sclavi, B., Ciandrini, L., Cosentino Lagomarsino, M.,
submitted to Nature Communications (Jan. 2025). 2025

Perturbing the catenoid: Stability and mechanical properties of nonaxisymmetric minimal surfaces
 Walzel, F., Requier, A., Boschi, K., Farago, J., **Fuchs, P.**, Thalmann, F., Drenckhan, W., Muller, P., Charitat, T.,
Phys. Rev. E 106 (1 July 2022) p. 014803. American Physical Society, 2022

IN PREPARATION

- Growth limits in G1 arrest and diluted cells**
 Fuchs P, Cosentino Lagomarsino M, Ciandrini L.
- Growth phases and limits in RNA Polymerase underexpression.**
 Fuchs P, Cosentino Lagomarsino M, Ciandrini L.

Interests

- Music** From listening to classics, Rock or playing myself (trombone), a day cannot pass without some music.
- Reading** From SF and fantasy books to classical works, I can pass days reading.
- Biking** I have always enjoyed biking for long trips out of the city.
- Fencing** I started fencing at 10, and it has remained an escape from stress for me all this time.

Languages

- French** Native proficiency
- Dutch** Native proficiency
- English** Professional proficiency (C1)
- German** Intermediate proficiency (B1)
- Spanish** Beginner proficiency (A2)
- Italian** Beginner proficiency (A2)

References available upon request.